

Uncertainty-Aware Symbolic Regression through Bayesian Support Selection

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The Sure Independence Screening and Sparsifying Operator (SISSO) framework is a powerful symbolic-regression method for extracting compact and interpretable descriptors from large non-linear feature spaces. However, standard SISSO is deterministic: it returns a single descriptor and point prediction, without quantifying uncertainty in descriptor selection, regression coefficients, or predictions. Here we introduce a probabilistic extension in which the sure independence screening (SIS) stage is kept deterministic to preserve scalability, while the sparsifying operator (SO) stage is reformulated as Bayesian inference over the SIS-screened support space. The resulting deterministic-SIS/Bayesian-SO framework yields posterior probabilities for competing descriptor supports, feature-inclusion probabilities, Bayesian-model-averaged predictions, and predictive credible intervals, while recovering the deterministic SO descriptor of standard SISSO in the maximum-a-posteriori limit. Applied to an X_2YZ Heusler-alloy magnetic-moment dataset, the approach gives modest improvements in five-fold cross-validation RMSE and near-nominal empirical coverage of the 95% predictive intervals. More importantly, the posterior exposes competing, physically related symbolic descriptor families that would appear artificially unique in a deterministic analysis. These results suggest that deterministic-SIS/Bayesian-SO can be used as an uncertainty-aware diagnostic extension of SISSO: a tool for assessing descriptor confidence, stability, and non-uniqueness in small-data materials regression problems.

Keywords: Symbolic regression, SISSO, Bayesian, Heusler alloys, magnetic moments

I. INTRODUCTION

Descriptor-based learning has become a central strategy in materials science because it balances predictive performance, data efficiency, and physical interpretability [1]. More broadly, the growth of materials informatics has also motivated automated approaches for extracting structured material-property information from the literature, including entity–relation extraction frameworks for constructing materials knowledge bases [2]. In many realistic materials problems, datasets are not large enough to justify purely black-box models, and the scientific credential of a model depends not only on its accuracy but also on whether it reveals an intelligible descriptor-property relationship. Symbolic regression is especially attractive in this setting because it searches for closed-form analytical expressions, typically involving only a few physically meaningful variables—that can be inspected, interpreted, and transferred by the respective domain experts.

Among symbolic-regression methods, SISSO is one of the most successful frameworks for navigating extremely large nonlinear feature spaces while retaining interpretability [1]. Starting from primary features, it constructs an enormous symbolic library through algebraic and functional operators, then applies sure independence screening (SIS) followed by sparse optimization (SO) to identify a low-dimensional descriptor. The practical impact of this framework is already evident across a broad range of materials problems [3–7]. SISSO-based descriptors have been used to guide the discovery of acid-stable

oxides for electrocatalysis, rationalize Curie-temperature trends in low-dimensional magnetic materials, and derive interpretable predictors for materials hardness [8–10]. These successes make SISSO particularly compelling for applications where one wants not just a predictor, but a compact analytical model that highlights the dominant physics.

The success of SISSO is closely linked to its deterministic two-stage structure. SIS aggressively reduces a combinatorially large feature library to a manageable subset, and SO selects a sparse descriptor from that reduced set. However, this deterministic pipeline does not directly provide posterior uncertainty for selected descriptors, regression coefficients and therefore the predictions. Such uncertainty is often critical in realistic small-data settings, where multiple highly correlated symbolic descriptors can explain the data similarly well.

Recent work has shown that SISSO can be embedded in Bayesian-optimization workflows for adaptive feature discovery, as in the SARBO framework of Nair et al. [11]. There, SISSO serves primarily as an adaptive representation-learning or feature-selection tool for the optimization surrogate. Here, we address a different problem: uncertainty quantification in symbolic descriptor selection itself. We therefore keep the SIS step deterministic for scalability, but replace the deterministic SO step by Bayesian inference over the SIS-screened symbolic supports.

In the present work, we develop a deterministic-SIS/Bayesian-SO variant of SISSO that preserves the practical scalability of the original method while adding uncertainty quantification at the descriptor-selection stage. Specifically, we retain SIS as a deterministic screening step, then perform Bayesian inference over

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the resulting screened support space to compute posterior support probabilities, feature-inclusion probabilities, Bayesian-model-averaged predictions, and predictive credible intervals. Applied to the X_2YZ Heusler-alloy [12] magnetic-moment dataset, this framework not only recovers the deterministic SO solution within the standard SISSO pipeline in the MAP limit, but also reveals competing descriptor families and improves cross-validation performance relative to that deterministic SIS+SO baseline. In this way, the method achieves the central goal of this work: uncertainty-aware symbolic regression without sacrificing the core computational efficiency of SISSO.

Because Bayesian SO requires a prior over symbolic supports, the prior should be chosen in a way that reflects the dataset, target property, and amount of available physical knowledge. In Bayesian inference, a prior distribution encodes information or preference about model parameters before observing the current data, and the likelihood updates this prior into a posterior distribution after the data are observed [13, 14]. In the present setting, the prior is placed on candidate descriptor supports: it can be chosen uniform, or it can weakly encode physically motivated preferences such as Slater–Pauling type electron-count structure. The Bayesian-SO framework requires a support prior, but the particular priors used here are not intrinsic to the method; they are illustrative choices designed to test generic, redundancy-aware, and weakly physics-guided support weighting. In this study, for the Heusler-alloy magnetic-moment data, we compare three support priors: (1) a uniform prior that treats all SIS-screened supports equally, (2) a relevance-diversity prior that favors individually relevant but mutually nonredundant features, and (3) a weak Slater–Pauling-inspired prior that encourages a two-dimensional descriptor organized around an electron-count backbone and a residual correction. These priors are specific choices for the present Heusler problem rather than universal prescriptions; other materials datasets may call for different weakly informative or physics-guided priors.

II. METHODOLOGY

A. Standard SISSO Framework

Following Ouyang *et al.* [1], SISSO starts from a set of primary features and recursively applies algebraic and functional operators to generate a very large candidate feature space for symbolic regression. For a dataset of N samples with targets y_i , the resulting symbolic library can be written as

$$\{\Phi_j(\mathbf{x})\}_{j=1}^M, \quad (1)$$

where M may be enormous. In a regression setting, the target vector is approximated as

$$\mathbf{y} \approx \Phi \mathbf{c}, \quad (2)$$

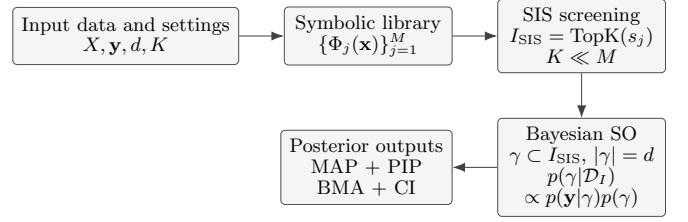


FIG. 1. Workflow of the deterministic-SIS/Bayesian-SO framework. SISSO constructs a symbolic feature library from the primary descriptors; deterministic SIS screening reduces this library to a tractable subspace; Bayesian SO evaluates sparse supports using marginal likelihoods and support priors. The posterior yields MAP descriptors, feature-inclusion probabilities, Bayesian-model-averaged predictions, and predictive credible intervals.

with sensing matrix $\Phi \in \mathbb{R}^{N \times M}$ and sparse coefficient vector $\mathbf{c}$. Direct optimization over all M candidates is generally intractable, so SISSO proceeds in two stages [1]. First, sure independence screening (SIS) ranks features according to their relevance to the target, or to the residuals from the previous descriptor dimension, and retains only a small screened subset. Second, the sparsifying operator (SO) performs an L_0 -style search within that reduced space to identify the best d -dimensional descriptor. The standard SISSO procedure therefore returns a single descriptor support and its associated regression coefficients. Throughout this manuscript, *standard SISSO* denotes this full deterministic SIS+SO pipeline, whereas *deterministic SO* denotes only the second-stage sparse search performed on a fixed SIS-screened feature set.

B. Deterministic SIS Stage

Let s_j denote the SIS relevance score for feature Φ_j , for example the absolute correlation with the target or with the residual at a given descriptor dimension. The screened index set is

$$\mathcal{I}_{\text{SIS}} = \text{TopK}(\{s_j\}_{j=1}^M; K), \quad |\mathcal{I}_{\text{SIS}}| = K, \quad K \ll M, \quad (3)$$

yielding a reduced design matrix

$$\Phi_{\mathcal{I}} \in \mathbb{R}^{N \times K}. \quad (4)$$

No probabilistic model is assigned to the discarded features. This is the central computational approximation that keeps the symbolic-regression problem tractable.

C. Bayesian Sparsifying Operator

Bayesian inference is performed only on the SIS-screened space. For a candidate support

$$\gamma \subset \{1, \dots, K\}, \quad |\gamma| = d, \quad (5)$$

we use the Gaussian linear model

$$\mathbf{y} = \Phi_{\mathcal{I}}^{(\gamma)} \beta_{\gamma} + \varepsilon, \quad \varepsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}), \quad (6)$$

where $\Phi_{\mathcal{I}}^{(\gamma)}$ denotes the screened columns indexed by γ . Active coefficients are assigned the prior

$$\beta_{\gamma} \sim \mathcal{N}(\mathbf{0}, \tau^2 \mathbf{I}). \quad (7)$$

Bayes' theorem gives

$$p(\gamma, \beta_{\gamma} \mid \mathcal{D}_{\mathcal{I}}) \propto p(\mathbf{y} \mid \gamma, \beta_{\gamma}, \sigma^2) p(\beta_{\gamma} \mid \tau^2) p(\gamma), \quad (8)$$

where $\mathcal{D}_{\mathcal{I}} = \{\Phi_{\mathcal{I}}, \mathbf{y}\}$. Integrating out coefficients yields the closed-form marginal likelihood

$$p(\mathbf{y} \mid \gamma) = \int p(\mathbf{y} \mid \gamma, \beta_{\gamma}, \sigma^2) p(\beta_{\gamma} \mid \tau^2) d\beta_{\gamma}. \quad (9)$$

The posterior support probability is therefore [15, 16],

$$p(\gamma \mid \mathcal{D}_{\mathcal{I}}) = \frac{p(\mathbf{y} \mid \gamma) p(\gamma)}{\sum_{\gamma'} p(\mathbf{y} \mid \gamma') p(\gamma')}. \quad (10)$$

In the implementation used for the cross-validation calculations, Bayesian inference is performed on the SISSO-screened feature values read from the standard SISSO output; no additional variance standardization of the symbolic features is applied. For each candidate support in a training fold, the active feature columns and the target are mean-centered when evaluating the marginal likelihood, which is equivalent to including an intercept. The intercept is restored as $\bar{y} - \bar{\phi}_{\gamma}^T \beta_{\gamma}$, and validation predictions are then reported on the original magnetic-moment scale using the corresponding screened feature columns. The parameter σ^2 is fixed globally to the full-data standard-SISSO RMSE squared for the corresponding descriptor dimension: $\sigma^2 = 0.26518$ for $d = 2$ and $\sigma^2 = 0.16937$ for $d = 3$. The coefficient-prior scale is fixed globally to the target variance divided by d , using $\text{var}(y) = 4.71578$, giving $\tau^2 = 2.35789$ for $d = 2$ and $\tau^2 = 1.57193$ for $d = 3$. These scales are not optimized on validation folds. Thus, the CV prediction errors are evaluated strictly on held-out targets, but the reported interval calibration should be interpreted conditional on these preset global variance scales rather than as the result of a fully nested variance-estimation procedure. The predictive variance for each support includes the observation-noise term and posterior coefficient covariance, and Bayesian model averaging incorporates support uncertainty through the posterior mixture over supports. Fold-specific quantities such as active-column means, target means, deterministic refits, and the Slater–Pauling residual prior are computed from training data only; validation targets are used only for scoring.

D. Support Priors and Predictions

For a fixed descriptor dimension d , we consider supports $\gamma \subset \{1, \dots, K\}$ with $|\gamma| = d$. The Bayesian-SO framework itself only requires that a prior $p(\gamma)$ be

specified on this support space. The uniform prior below is the generic default, while the relevance-diversity and Slater–Pauling priors are optional examples used to probe how redundancy-aware and weakly physics-guided support weighting affect the Heusler-alloy results. For the two-dimensional Heusler descriptor we compare all three priors, and for the three-dimensional benchmark we restrict the comparison to the generic uniform prior and the optional relevance-diversity prior.

The first is a *uniform prior*, which assigns equal probability to every admissible support and serves as the method's generic, dataset-agnostic choice:

$$p_{\text{unif}}(\gamma) = \begin{cases} \binom{K}{d}^{-1}, & |\gamma| = d, \\ 0, & \text{otherwise.} \end{cases} \quad (11)$$

This prior expresses no preference among screened features beyond the deterministic SIS reduction itself. It should be noted here that although the uniform prior assigns equal prior probability to all SIS-screened supports of size d , Bayesian SO is not equivalent to deterministic SO. Deterministic SO selects the single support with the best optimization criterion, whereas uniform-prior Bayesian SO converts the marginal likelihoods of all screened supports into posterior probabilities. Thus, the uniform prior removes prior preference among supports but still allows the data likelihood to quantify support competition, descriptor non-uniqueness, and model-averaged prediction uncertainty.

The second is the *relevance-diversity prior*, which favors supports containing individually relevant SIS features while discouraging redundant, highly correlated descriptor components:

$$p_{\text{RD}}(\gamma) \propto \exp \left[\lambda_{\text{SIS}} \sum_{k \in \gamma} \log s_k - \lambda_{\text{corr}} \sum_{\substack{k, l \in \gamma \\ k < l}} |\text{corr}(\Phi_k, \Phi_l)| \right], \quad (12)$$

$|\gamma| = d,$

and $p_{\text{RD}}(\gamma) = 0$ for supports of any other size. Here s_k is the SIS score of screened feature k , $\lambda_{\text{SIS}} \geq 0$ controls how strongly the prior rewards high-scoring features, and $\lambda_{\text{corr}} \geq 0$ penalizes supports whose members are strongly correlated with one another. The first term therefore rewards features with high SIS scores (s_k), while the second term penalizes internal redundancy.

The reported relevance-diversity calculations use $\lambda_{\text{SIS}} = 1$ and $\lambda_{\text{corr}} = 5$. These values were chosen as weak diagnostic weights: the SIS term preserves the deterministic screening order, while the correlation term discourages nearly collinear supports without excluding them. The prior is evaluated as an unnormalized log prior; posterior probabilities are normalized over the enumerated support set through the evidence denominator in the support-posterior expression above. Because all enumerated supports have the same fixed size d , any support-independent prior normalizing constant cancels in the posterior.

E. Slater-Pauling prior for two-dimensional Heusler descriptors

For full-Heusler alloys X_2YZ , the total valence electron count is [17, 18]

$$N_v = 2N_X + N_Y + N_Z. \quad (13)$$

For full-Heusler alloys, the Slater-Pauling rule is commonly written as $M_t = Z_t - 24$, reflecting the occupation of 12 minority-spin states. We therefore keep the electron-count slope fixed at unity and allow only family-dependent offsets for Co-, Fe-, and Ni-based compounds. In each cross-validation fold, the Slater-Pauling baseline is fit on the training data only as

$$M_i^{\text{SP}} = (N_{v,i} - 24) + b_{X_i}, \quad (14)$$

where $X_i \in \{\text{Co}, \text{Fe}, \text{Ni}\}$ denotes the X-site family and

$$b_X = \langle M_i - (N_{v,i} - 24) \rangle_{i \in X, \text{ train}}. \quad (15)$$

The residual is then

$$r_i^{\text{SP}} = M_i - M_i^{\text{SP}}. \quad (16)$$

For a two-dimensional support γ , the Slater-Pauling prior is

$$p_{\text{SP}}(\gamma) \propto \exp \left[\lambda_{N_v} Q_{N_v}(\gamma) + \lambda_{\text{res}} Q_{\text{res}}(\gamma) + \lambda_{\text{split}} B_{\text{split}}(\gamma) - \lambda_{\text{corr}} Q_{\text{corr}}(\gamma) \right], \quad (17)$$

with

$$Q_{N_v}(\gamma) = \max_{k \in \gamma} |\text{corr}(\Phi_k, N_v)|, \quad (18)$$

$$Q_{\text{res}}(\gamma) = \max_{k \in \gamma} |\text{corr}(\Phi_k, r^{\text{SP}})|, \quad (19)$$

$$Q_{\text{corr}}(\gamma) = \sum_{\substack{k, l \in \gamma \\ k < l}} |\text{corr}(\Phi_k, \Phi_l)|. \quad (20)$$

Here $B_{\text{split}}(\gamma) = 1$ when the feature most aligned with N_v differs from the feature most aligned with r^{SP} , and is zero otherwise. This prior is an optional Heusler-specific example rather than a required component of Bayesian SO. It is used only for $d = 2$ because a two-dimensional support has a clear interpretation as an electron-count backbone plus one symbolic correction term. The prior we consider is weak: it does not impose the ideal Slater-Pauling rule as a hard constraint, but instead organizes the posterior around an electron-count *plus* residual-correction interpretation. The Slater-Pauling prior is chosen to be weak because the actual Slater-Pauling rule provides a physically motivated electron-counting reference for full-Heusler alloys [17], but it is not expected to hold exactly across the present chemically diverse dataset. Deviations can arise from the loss of ideal half-metallicity, changes in hybridization, chemical-family dependence, and finite residual contributions beyond simple valence counting. The prior therefore does not impose

the Slater-Pauling rule as a hard constraint. Instead, it mildly favors supports that separate an electron-count-like backbone from a residual correction coordinate, while allowing the likelihood to determine the posterior weight of each symbolic descriptor.

For the Slater-Pauling prior we use $\lambda_{N_v} = 0.5$, $\lambda_{\text{res}} = 1.0$, $\lambda_{\text{split}} = 0.25$, and $\lambda_{\text{corr}} = 0.5$. The smaller correlation penalty relative to the relevance-diversity prior is intentional: the Slater-Pauling prior is meant to organize supports into an electron-count-like coordinate plus a residual-correction coordinate, not to make decorrelation the dominant prior preference. As for the relevance-diversity prior, the Slater-Pauling prior is normalized only through the posterior normalization over the enumerated supports.

Posterior inclusion probabilities quantify descriptor stability:

$$P(k \in \gamma \mid \mathcal{D}_{\mathcal{I}}) = \sum_{\gamma \ni k} p(\gamma \mid \mathcal{D}_{\mathcal{I}}). \quad (21)$$

Predictions are then obtained by Bayesian model averaging over all screened supports,

$$p(y_* \mid \mathbf{x}_*, \mathcal{D}_{\mathcal{I}}) = \sum_{\gamma} p(\gamma \mid \mathcal{D}_{\mathcal{I}}) p(y_* \mid \mathbf{x}_*, \gamma, \mathcal{D}_{\mathcal{I}}), \quad (22)$$

where each term $p(y_* \mid \mathbf{x}_*, \gamma, \mathcal{D}_{\mathcal{I}})$ is the predictive distribution conditioned on one candidate support. The resulting mixture yields both a posterior-mean prediction and predictive credible intervals. In the MAP limit, this formulation reduces to deterministic SO selection of the single highest-posterior support on the SIS-screened space, which is the second stage of standard SISSO.

III. DATASET: X_2YZ HEUSLER ALLOYS

The dataset used for the SISSO analysis consists of full-Heusler alloys with nominal composition X_2YZ , where X is Co, Fe, or Ni. The compound list and associated materials information were taken from the University of Alabama Heusler Alloy Database [19]. The dataset contains 88 compounds: 38 Co-based, 32 Fe-based, and 18 Ni-based alloys. For these X_2YZ compounds, the two X atoms occupy equivalent positions, while Y and Z represent the other distinct sublattices.

For each of the X, Y, and Z sites, the descriptor set includes the first ionization energy (I_E), electronegativity (X_P), covalent radius (R_c), number of valence electrons (N), atomic volume (V), and number of d electrons (N_d). This yields 17 site-resolved primary features (we drop N_d for Z as they are all zeros) and provides a compact representation of the chemical size, electronic structure, and valence characteristics of the constituent elements.

Figure 2 summarizes the distribution of magnetic moments and the Pearson-correlation structure of the primary descriptors. The magnetic-moment histogram

[Fig. 2(a)] shows a clear separation among the three X-site families. Co-based alloys typically span 1 to 6 $\mu_B/f.u.$ (median 3.58 $\mu_B/f.u.$). Fe-based alloys show a broader distribution, including both nearly non-magnetic and high-moment compounds (up to 6.60 $\mu_B/f.u.$). The Ni-based subset is dominated by zero-moment compounds.

The Pearson correlation heatmap [Fig. 2(b)] reveals substantial multicollinearity among the site-resolved elemental descriptors. Clear blocks of positive and negative correlation appear among chemically related variables across the X, Y, and Z sublattices, especially for descriptors associated with valence, atomic size, and d -electron character. This inter-feature correlation is important for symbolic regression because many candidate expressions are built from inputs that are not statistically independent.

The direct Pearson correlations with the magnetic moment, are dominated by Y-site descriptors. The strongest positive correlations are N_Y (0.873), $I_{E,Y}$ (0.834), and $X_{P,Y}$ (0.629), whereas the strongest negative correlations are $N_{d,Y}$ (-0.859), $R_{c,Y}$ (-0.845), and V_Y (-0.659). The strong negative correlation of $N_{d,Y}$ with magnetic moment may therefore reflect how progressive filling of the Y-site d shell changes the X-Y hybridization pattern [20] and the occupation of spin-resolved d -derived states, rather than a simple one-variable causal effect. In contrast, the reported X- and Z-site descriptors are noticeably weaker, with the largest magnitude among them coming from $I_{E,X}$ (0.490). Taken together, the heatmap and target-correlation ranking indicate that the Y sublattice carries the most direct correlation for the magnetic moment, while the full descriptor set remains strongly correlated internally.

A. Standard SISSO setup

The standard SISSO calculations were carried out using the same primary feature set for both descriptor dimensions. The regression type was set to single-task regression, and all calculations used the full set of $N = 88$ samples. The symbolic feature space was generated from the primary features using the operators

$$(+), (-), (*), (/), (\cdot)^{-1}, (\cdot)^2,$$

with maximum feature complexity of five. The feature-value range was restricted by $1.0e-3$ and $1.0e5$. Dimensional consistency was controlled through the SISSO unit groups.

The sparsifying-operator stage used ℓ_0 optimization, with an intercept included and RMSE as the selection metric.

The SIS screening depth was specified per descriptor dimension. For the two-dimensional search, 200 ranked features were retained in each SIS step, and the union of the two screened sets produced a Bayesian support space with $K = 400$ candidate features. For the three-dimensional search, 100 ranked features were retained in

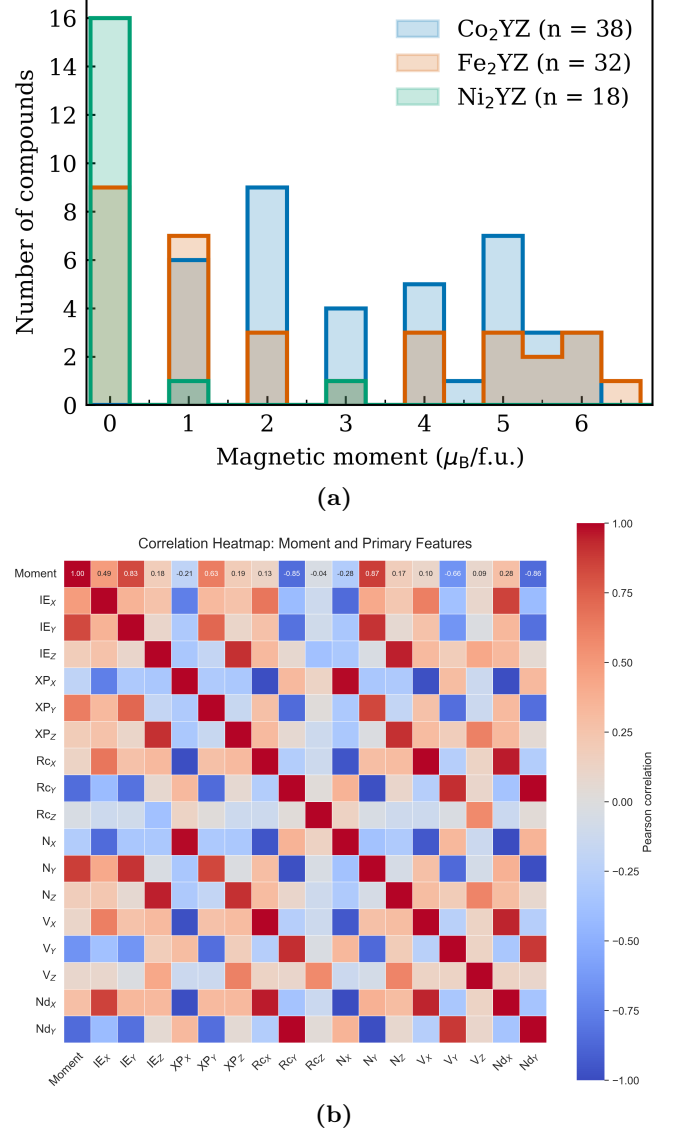


FIG. 2. Summary of the X_2YZ Heusler alloy dataset. (a) Distribution of magnetic moments for Co-, Fe-, and Ni-based alloys. (b) Pearson correlation heatmap among the 18 site-resolved elemental descriptors used as SISSO inputs.

each SIS step, giving $K = 300$ candidate features in the union of the three screened sets. We therefore distinguish the per-step SIS retention depth from K , the total screened feature count used for Bayesian enumeration. The resulting support spaces, $\binom{400}{2} = 7.98 \times 10^4$ and $\binom{300}{3} = 4.4551 \times 10^6$, retain the full deterministic SISSO screened unions from the corresponding runs while remaining tractable for exact posterior enumeration.

IV. RESULTS AND DISCUSSION

In this section, Bayesian model averaging (BMA) denotes posterior-weighted averaging of predictions over

candidate supports, and posterior inclusion probability (PIP) denotes the posterior probability that a given screened feature appears in the descriptor support.

The SISO calculations use a rung-three symbolic library with addition, subtraction, multiplication, division, inverse, and square operators. The resulting feature space contains 2.425×10^8 candidate expressions before SIS screening. We test two descriptor dimensions. For $d = 2$, SIS retains $K = 400$ screened features, giving $\binom{400}{2} = 7.98 \times 10^4$ candidate supports. For $d = 3$, SIS retains $K = 300$ features, giving $\binom{300}{3} = 4.4551 \times 10^6$ supports. These support spaces are still small enough for exact Bayesian enumeration after deterministic SIS.

On the full training set, standard SISO selects the two-dimensional support (8, 203) with RMSE 0.5150, and the three-dimensional support (8, 101, 203) with RMSE 0.4115. Feature 8,

$$\phi_8 = ((V_X)^2 N_X) / (N_X + N_Y + N_Z), \quad (23)$$

appears in both descriptors and remains the most stable Bayesian feature below. It combines the X-site atomic volume and valence count with the total valence-electron scale, consistent with the importance of electron counting for Heusler magnetic moments.

We emphasize that feature 8 should not be interpreted as a *kind of alternative* to the standard Slater-Pauling rule. The latter is an electron-counting relation based on minority-spin band filling and involves the full-Heusler valence count $N_v = 2N_X + N_Y + N_Z$. By contrast, feature 8 is a SISO-selected symbolic coordinate with composite units and fitted regression coefficients. Its repeated posterior selection should therefore be interpreted more conservatively: it indicates that X-site size and valence information, normalized by an electron-count-like scale, provide a statistically stable descriptor component for this dataset. This is physically plausible because X-site chemistry can influence the lattice scale, d -band bandwidth, hybridization, and magnetic polarization, but the precise V_X^2 dependence is best viewed as an empirical symbolic representation rather than a microscopic law. The appearance of V_X -dependent terms is qualitatively consistent with itinerant-magnetism arguments, since increasing atomic volume can reduce d -band width and enhance the density of states entering the Stoner criterion [21]. However, the specific V_X^2 dependence selected by SISO should not be interpreted as a microscopic consequence of the Stoner criterion. A simple bandwidth-scaling argument would suggest a volume dependence closer to $V^{5/3}$, so the V_X^2 factor is best viewed as an empirical symbolic proxy for X-site size, bandwidth, and hybridization effects within the present dataset.

Feature indices are inherited from the SISO-generated symbolic library and are not constrained to respect exact stoichiometric electron counting unless such constraints are explicitly imposed. Therefore, feature 8 should be distinguished from the canonical Slater-Pauling count $N_v = 2N_X + N_Y + N_Z$. As in many SISO applications, symbolic coordinates may carry composite physical

TABLE I. Five-fold cross-validation comparison between the deterministic SIS+SO baseline of standard SISO and deterministic-SIS/Bayesian-SO on the Heusler-alloy magnetic-moment dataset. The Bayesian entry is the posterior-mean predictor. The deterministic baseline is independent of the Bayesian support prior: $\text{RMSE}_{\text{det}} = 0.5609$ for $d = 2$ and $\text{RMSE}_{\text{det}} = 0.4515$ for $d = 3$. Coverage is the empirical coverage of the Bayesian 95% predictive interval.

d	Prior	K	Supports	Bayesian RMSE	Coverage
2	uniform	400	7.98×10^4	0.5314	0.9549
2	relevance-diversity	400	7.98×10^4	0.5334	0.9549
2	Slater-Pauling	400	7.98×10^4	0.5317	0.9549
3	uniform	300	4.4551×10^6	0.4467	0.9660
3	relevance-diversity	300	4.4551×10^6	0.4501	0.9660

units; the fitted regression coefficients restore the target units. We therefore interpret the selected expressions as compact empirical descriptors rather than parameter-free analytical laws.

The second coordinate in the $d = 2$ descriptor belongs to the 201/203 family,

$$\phi_{201} = ((N_X V_Y) / (N_Z V_X)) - (N_{d,Y} / N_{d,X}), \quad (24)$$

and

$$\phi_{203} = ((N_X V_Y) / V_X) - ((N_Z N_{d,Y}) / N_{d,X}). \quad (25)$$

Chemically, these expressions are not arbitrary combinations of variables: they compare Y-site valence-volume contributions against Z-site valence and X-site d -electron filling. This is a natural structure for full-Heusler compounds, where the Y atom strongly tunes the magnetic moment through its valence and d -state occupancy, while the Z atom acts more as a main-group electron reservoir. The posterior competition between features 201 and 203 therefore indicates that the data robustly favor a particular physical motif—a valence/volume contrast modulated by d -electron mismatch, even when the exact algebraic form is not unique.

Table I and Fig. 3 show that Bayesianizing the SO stage does not degrade the deterministic SIS+SO baseline of standard SISO. For $d = 2$, Bayesian model averaging reduces the average validation RMSE from the deterministic baseline 0.5609 to 0.5314 under the uniform prior, 0.5334 under the relevance-diversity prior, and 0.5317 under the Slater-Pauling prior. These three Bayesian values are practically indistinguishable. Thus, the validation improvement is not primarily a consequence of prior engineering; rather, Bayesian SO replaces a single deterministic SO descriptor by a posterior distribution over a small family of statistically competitive symbolic descriptors. The uniform prior already captures this non-uniqueness, while the relevance-diversity and Slater-Pauling priors provide alternative, physically interpretable organizations of essentially the same posterior family.

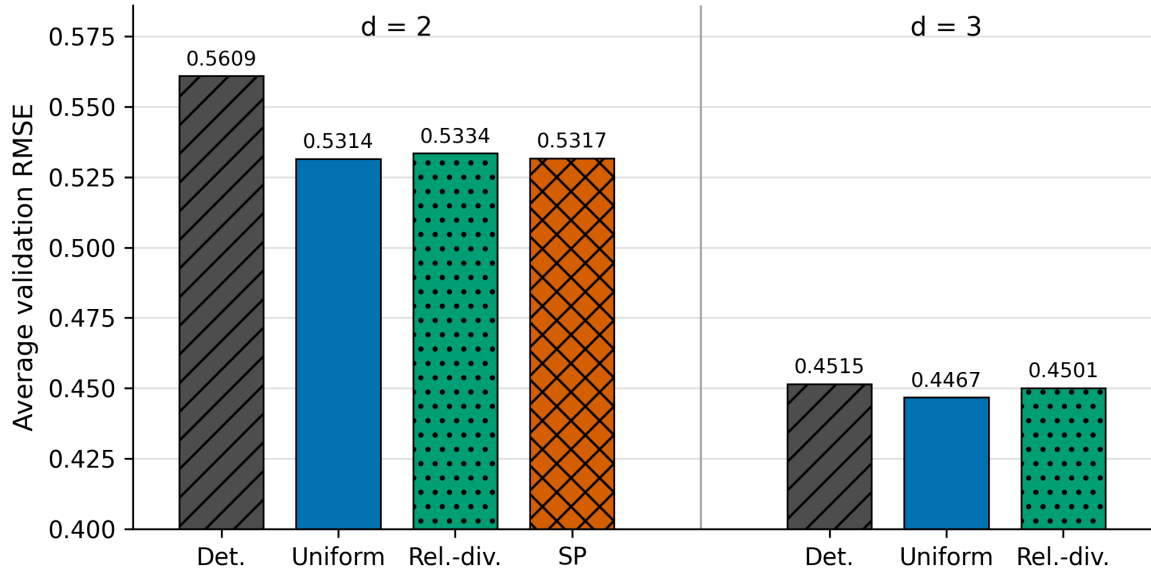


FIG. 3. Average validation RMSE for the deterministic SIS+SO baseline of standard SISSO and Bayesian posterior-mean predictions. The deterministic bar is the shared baseline for each descriptor dimension; the Bayesian support prior affects only the Bayesian-SO calculation. For $d = 2$, the three Bayesian priors give nearly identical RMSEs, showing that prior choice is not the dominant source of the validation improvement.

The improvement is largest in fold 3, where the deterministic SO stage selects the different support (4, 326), while the Bayesian MAP support remains in the recurring (8, 201) or (8, 203) family. This is precisely the regime where a single deterministic L_0 optimum is fragile and model averaging is useful.

For $d = 3$, both methods operate on a lower error scale, and the Bayesian gain is smaller. The uniform-prior Bayesian RMSE is 0.4467, compared with 0.4515 for the deterministic SIS+SO baseline; the relevance-diversity result is 0.4501. This behavior is expected. The third descriptor component already stabilizes the deterministic model, with fold MAP supports concentrated near (8, 101, 203) and (8, 103, 203), leaving less residual descriptor ambiguity for Bayesian averaging to exploit. The Slater–Pauling prior is not applied to $d = 3$, because its backbone-plus-correction interpretation is more relevant to two-dimensional supports.

The Bayesian 95% predictive intervals are well calibrated. The empirical coverage is 0.9549 for all three two-dimensional priors and 0.9660 for both three-dimensional priors, close to the nominal value. Fold 3 is again the most difficult split, with coverage 0.8889, but the average coverage remains near 95%. This is a central practical advantage over the deterministic SIS+SO pipeline: the standard SISSO descriptor gives a point prediction, whereas Bayesian SO gives a point prediction and an uncertainty estimate tied directly to coefficient and support uncertainty.

TABLE II. Sensitivity of the two-dimensional uniform-prior CV posterior to the screened-space size K . PIPs are five-fold averages.

K	Supports	Bayesian RMSE	Coverage	PIP(8)	PIP(201)	PIP(203)
200	2.01×10^4	0.5206	0.9549	0.9810	0.0000	0.9999
300	4.485×10^4	0.5313	0.9549	0.9721	0.5348	0.3755
400	7.98×10^4	0.5314	0.9549	0.9717	0.5343	0.3752

A. Sensitivity checks

Table II summarizes a two-dimensional K -sensitivity check using the uniform prior and the same five CV splits. The $K = 300$ and $K = 400$ results are essentially identical: the average RMSE, coverage, MAP supports, and leading PIPs all remain unchanged within numerical precision. The smaller $K = 200$ screen is qualitatively different because feature 201 is not present in that truncated support space; the posterior therefore assigns nearly all second-coordinate probability to feature 203. This result clarifies that the main posterior conclusion is stable once the competing 201/203 descriptor family is included in the screened set. A $K = 600$ calculation was not possible from the present deterministic SISSO outputs because the saved SIS-screened union contains only 400 features for $d = 2$; it would require repeating the deterministic SIS stage with a larger per-step screening depth.

Additional variance-scale sensitivity and computational-cost details are reported in the Appendix. In brief, the dominant conclusions are unchanged over the tested σ^2 and τ^2 ranges: the two-dimensional MAP

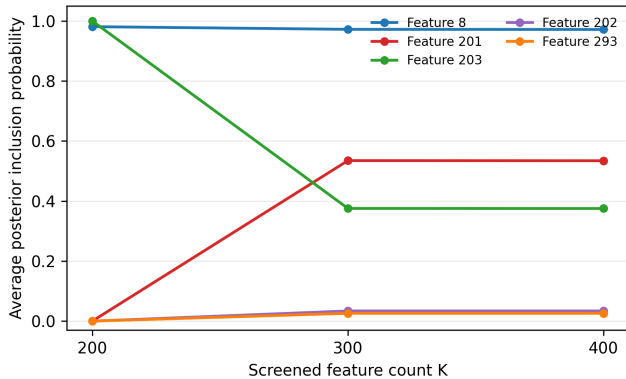


FIG. 4. Two-dimensional uniform-prior PIP stability with respect to the screened-space size K . Feature 201 is absent from the $K = 200$ truncated screen, so the posterior concentrates on feature 203. Once both members of the 201/203 family are available ($K = 300$ and $K = 400$), the leading PIPs are essentially unchanged.

support remains within the (8, 201)/(8, 203) family, while the three-dimensional MAP support remains fixed at (8, 101, 203) for both the uniform and relevance-diversity priors.

Finally, the present calculations use centered but not variance-standardized symbolic features, and they do not implement a Zellner g -prior or column-scaled τ_k [22]. Such scale-adaptive priors are natural extensions because SISSO expressions can have very different numerical scales. We also did not perform bootstrap-SISSO, BIC-weighted deterministic-support averaging, or spike-and-slab regression baselines in this work. The comparisons reported here are therefore against the deterministic SIS+SO baseline of standard SISSO and the family-offset Slater–Pauling reference; broader UQ-baseline comparisons remain useful future benchmarks.

B. Posterior feature stability and Slater–Pauling-prior comparison

As already mentioned, the Slater–Pauling prior is somewhat weak. Its average RMSE, 0.5317, is nearly identical to the uniform-prior value 0.5314, indicating that the fold likelihood still controls the posterior and that the prior does not force the solution away from the data-dominated support family. Its purpose is therefore not to outperform the uniform prior numerically, but to test whether a weak support bias can preserve the same predictive accuracy while making the posterior organization more interpretable. In this sense, it should be treated as a diagnostic or interpretability prior, not as a predictive-performance prior. The MAP supports under this prior remain in the same physically interpretable family as the uniform and relevance-diversity analyses: four of the five folds select (8, 201), and one

selects (8, 203). The family offsets fitted on the training folds are small for Co-based compounds (-0.014 to $0.007 \mu_B$), moderately positive for Fe-based compounds (0.167 to $0.423 \mu_B$), and strongly negative for Ni-based compounds (-3.478 to $-3.073 \mu_B$). The family-corrected Slater–Pauling baseline has an average validation RMSE of 0.6525, so it captures the electron-count trend but remains less accurate than the Bayesian symbolic descriptor. The residual-correlation and split-role terms are therefore useful as interpretability biases: they favor a descriptor organization with one electron-count backbone coordinate and one correction coordinate without imposing the ideal Slater–Pauling rule as a hard constraint.

The feature-stability plots in Fig. 6 show that the posterior is not simply smoothing predictions; it identifies a physically interpretable descriptor family. In $d = 2$, feature 8 has average inclusion probability 0.972 with the uniform prior and 0.958 with the relevance-diversity prior. The second coordinate is shared mainly between features 201 and 203, with uniform-prior inclusion probabilities 0.534 and 0.375. Thus, the deterministic descriptor selects one member of the valence–volume-contrast family defined above, but the posterior correctly reports that the data do not uniquely resolve the choice.

The same pattern persists in $d = 3$. For the uniform prior, the leading inclusion probabilities are 0.990, 0.757, and 0.670 for features 8, 101, and 203, respectively; the relevance-diversity prior gives the same ordering with probabilities 0.979, 0.746, and 0.657. Competing features 103, 202, and 201 retain non-negligible posterior weight. Thus the uncertainty is localized within chemically similar symbolic alternatives rather than spread uniformly over the screened feature space.

The physical interpretation of the three-dimensional descriptor is therefore hierarchical rather than completely new. The $d = 2$ backbone already captures the dominant balance between the magnetic X-site framework and the Y/Z-dependent electronic contrast. The additional third coordinate in the $d = 3$ family, represented most often by feature 101 and sometimes by chemically similar alternatives such as 103 or 202, acts as a refinement term that resolves finer differences among alloys with otherwise similar electron-count and volume balance. In other words, $d = 3$ does not replace the chemical picture learned at $d = 2$; it sharpens it by accounting for secondary variations in local electronic structure. This also explains why moving from $d = 2$ to $d = 3$ lowers the overall error but reduces the relative benefit of Bayesian averaging: once the dominant and subleading chemical trends are both represented, there is less residual ambiguity left for model averaging to exploit.

The uniform, relevance-diversity, and Slater–Pauling priors lead to nearly the same two-dimensional conclusions. The relevance-diversity prior slightly redistributes posterior mass by penalizing strongly correlated supports, while the Slater–Pauling prior mildly favors one electron-count-aligned feature together with one residual-correction feature. In both cases, the main descriptor

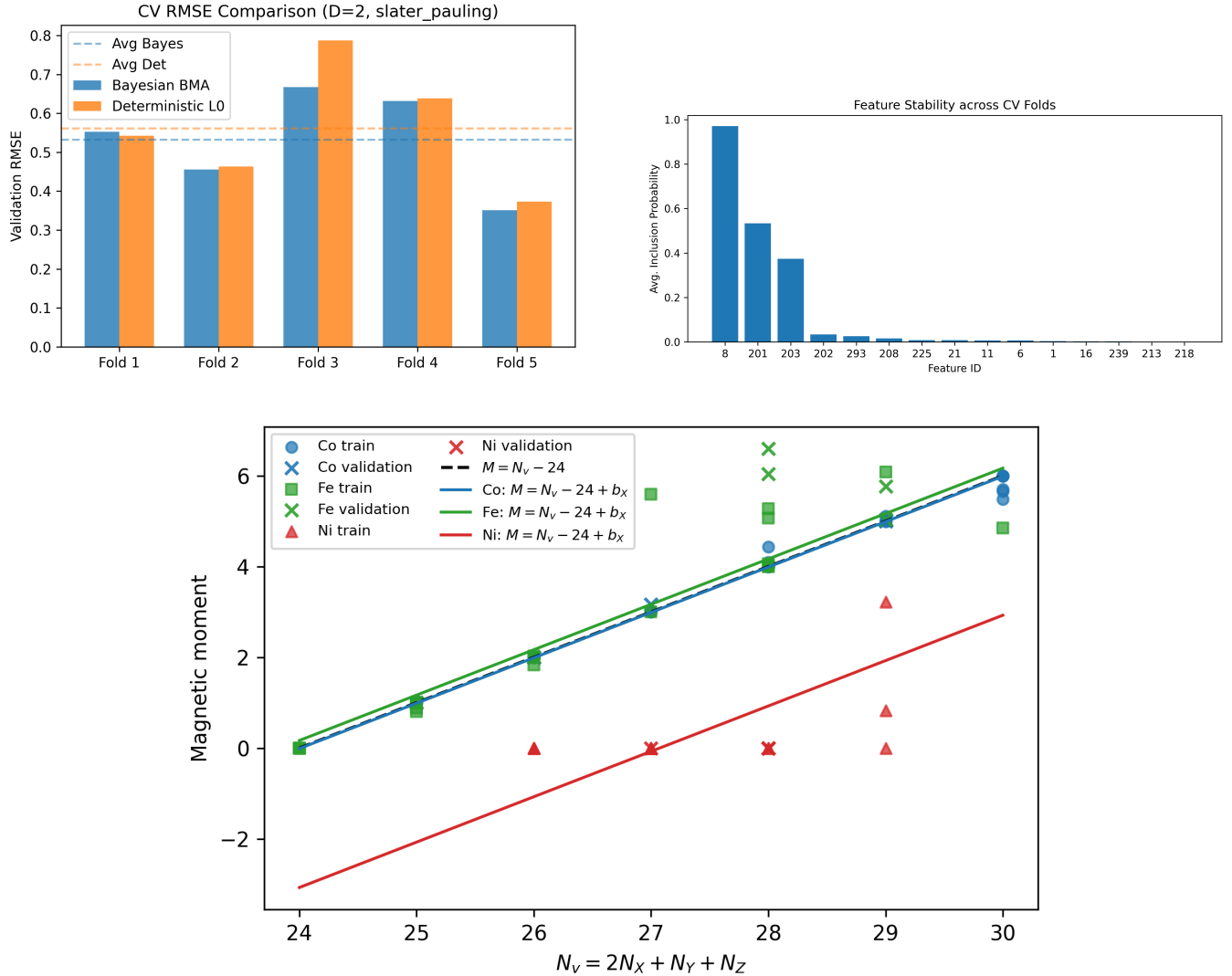


FIG. 5. Diagnostics for the two-dimensional Slater–Pauling prior. Top left: fold-resolved validation RMSE. Top right: average posterior inclusion probabilities. Bottom: magnetic moment versus valence count with the ideal line $M = N_v - 24$ and fold-1 family-offset Slater–Pauling baselines; validation points in that fold are shown but were not used to fit the offsets.

family, the validation RMSE, and the predictive coverage remain essentially unchanged. This robustness is important: it indicates that the posterior is controlled primarily by the data and the SIS-screened symbolic space, not by an overly strong prior. The Slater–Pauling prior is therefore best viewed as an interpretability-preserving diagnostic prior rather than an accuracy-enhancing or predictive-performance prior.

V. CONCLUSION

The deterministic-SIS/Bayesian-SO construction is a conservative extension of standard deterministic SISSO. It keeps the scalable deterministic screening step intact, recovers the deterministic SO support of standard

SISSO as a MAP limit, and restricts Bayesian inference to the tractable SIS-screened support space. In five-fold cross-validation on the Heusler-alloy magnetic-moment dataset, Bayesian model averaging modestly lowers the average validation RMSE for both $d = 2$ and $d = 3$ descriptors, while the nearly identical $d = 2$ results under the uniform, relevance-diversity, and Slater–Pauling priors show that the effect is not driven by prior-specific performance gains. Bayesian SO also supplies calibrated predictive intervals and posterior inclusion probabilities. More importantly, the posterior reveals descriptor non-uniqueness that is hidden when the deterministic SO stage returns only a single support: a selected standard-SISSO descriptor may be only one representative of a broader family of statistically competitive and chemically related symbolic expressions. The selected expressions

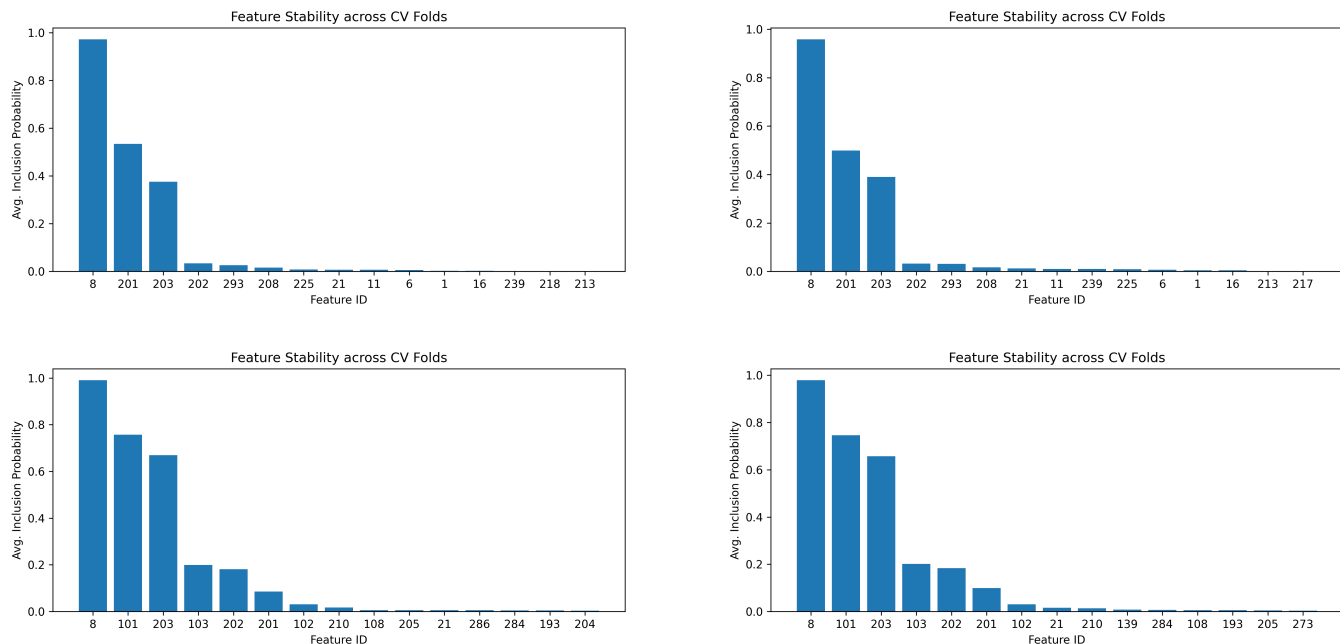


FIG. 6. Posterior feature-inclusion probabilities averaged over the five cross-validation folds. Each bar gives the average posterior probability that a screened symbolic feature is included in the Bayesian-SO support. The four panels correspond to different descriptor dimensions and support priors: top left: $d=2$, uniform prior; top right: $d=2$, relevance-diversity prior; bottom left: $d=3$, uniform prior; bottom right: $d=3$, relevance-diversity prior. Feature 8 has the largest inclusion probability in all cases, while features 101, 201, 202, and 203 form the main competing descriptor family.

should therefore be interpreted as posterior-stable symbolic descriptors whose chemical plausibility can be assessed a posteriori, rather than as parameter-free microscopic laws. The Slater–Pauling prior is best viewed as a diagnostic prior that organizes this family in an electron-count-based way, not as a route to improved accuracy. This makes the approach useful not only as a predictor, but also as a diagnostic tool for assessing descriptor stability in small-data materials problems, where correlated features and nearly degenerate descriptors are common.

Appendix A: Additional Sensitivity and Computational Details

We tested the sensitivity of the full-data posterior mean to the variance scales σ^2 and τ^2 . For $d=2$, the grid used $\sigma^2 \in \{0.1326, 0.1989, 0.2652, 0.3315, 0.3978, 0.5304\}$ and $\tau^2 \in \{0.5895, 1.1789, 2.3579, 4.7158, 9.4316\}$. Across these 30 combinations, the two-dimensional uniform-prior RMSE varied only from 0.5079 to 0.5148, and the MAP support remained within the (8,201)/(8,203) family. The relevance-diversity result was similarly stable, with RMSE from 0.5079 to 0.5164 and the same MAP-support family. For $d=3$, the grid used $\sigma^2 \in \{0.0847, 0.1270, 0.1694, 0.2117, 0.2541, 0.3387\}$ and $\tau^2 \in \{0.3930, 0.7860, 1.5719, 3.1439, 6.2877\}$. The three-dimensional MAP support stayed fixed at (8,101,203) for all 30 grid points under both the uniform and relevance-

diversity priors; the RMSE ranges were 0.3994–0.4120 and 0.4017–0.4119, respectively. These grids were not used to tune the CV results and do not constitute a nested CV optimization; they are reported only as scale-sensitivity diagnostics.

The enumeration cost is controlled primarily by $\binom{K}{d}$. On the workstation used for these calculations (Intel Xeon Gold 6240R, 48 logical CPUs, 30 GiB RAM; Python 3.9.12, NumPy 1.26.4, scikit-learn 1.4.2, joblib 1.4.2), a single full-data $d=2$, $K=400$ uniform-prior enumeration over 7.98×10^4 supports took 2.6 s wall time and 74 MB maximum resident memory. A single $d=3$, $K=300$ enumeration over 4.4551×10^6 supports took 27.2s and 0.76 GB for the uniform prior, and 31.8s and 0.76 GB for the relevance-diversity prior. Five-fold CV multiplies this cost by five for each prior, with small additional overhead for deterministic refits and plotting. The deterministic full-data SISSO runs took 227s for $d=2$ and 292s for $d=3$ on the same saved calculations. In the present implementation, exact enumeration at $d=3$, $K \approx 300$ is practical, while $d=4$ or substantially larger K would require either more aggressive screening, streaming posterior accumulation, stochastic support sampling, or a different approximation.

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